

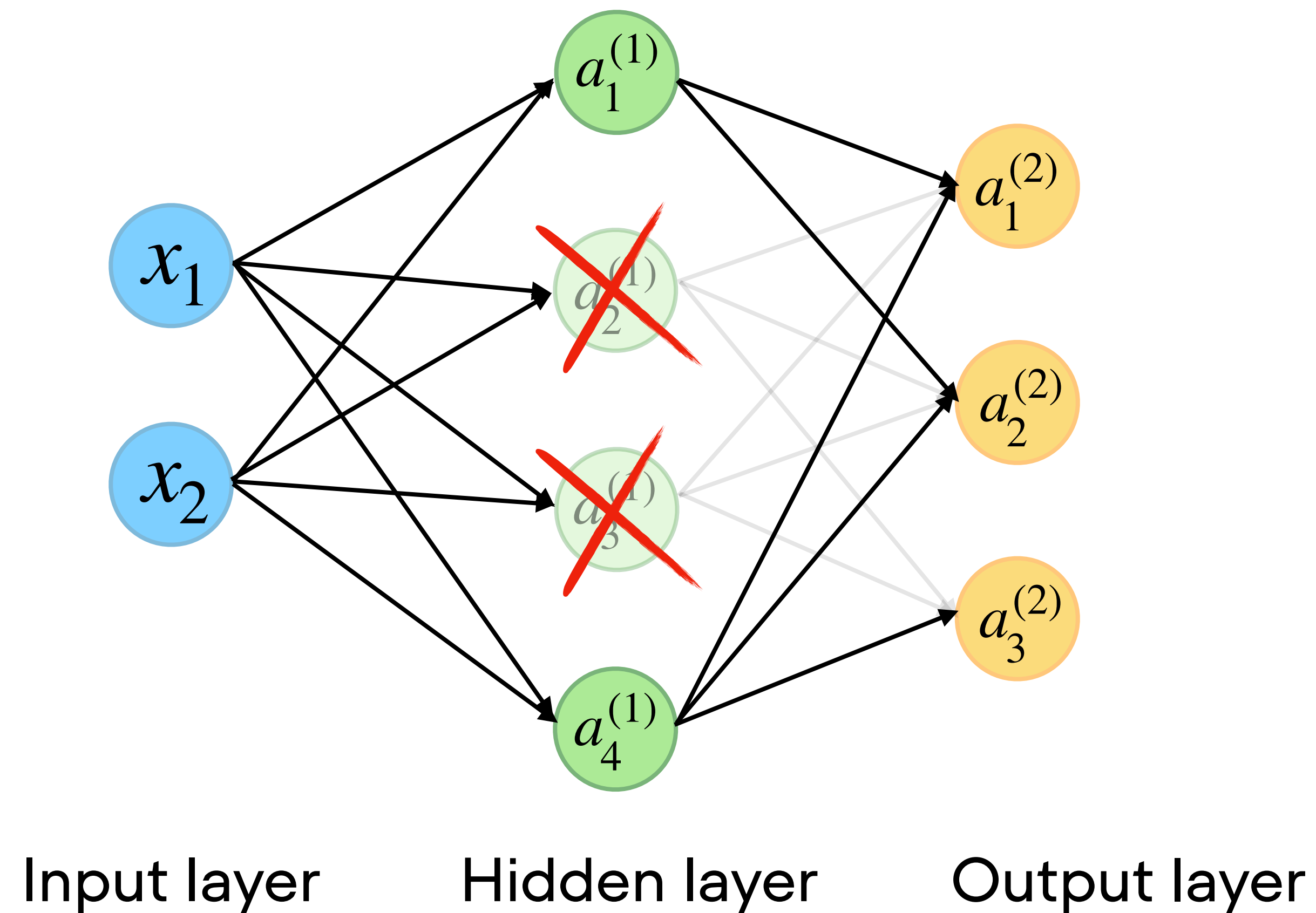
6.7

Reducing Overfitting With Dropout

Part 2: Applying Dropout During Inference

Sebastian Raschka and the Lightning AI Team

Dropout in a nutshell: randomly drop nodes during training

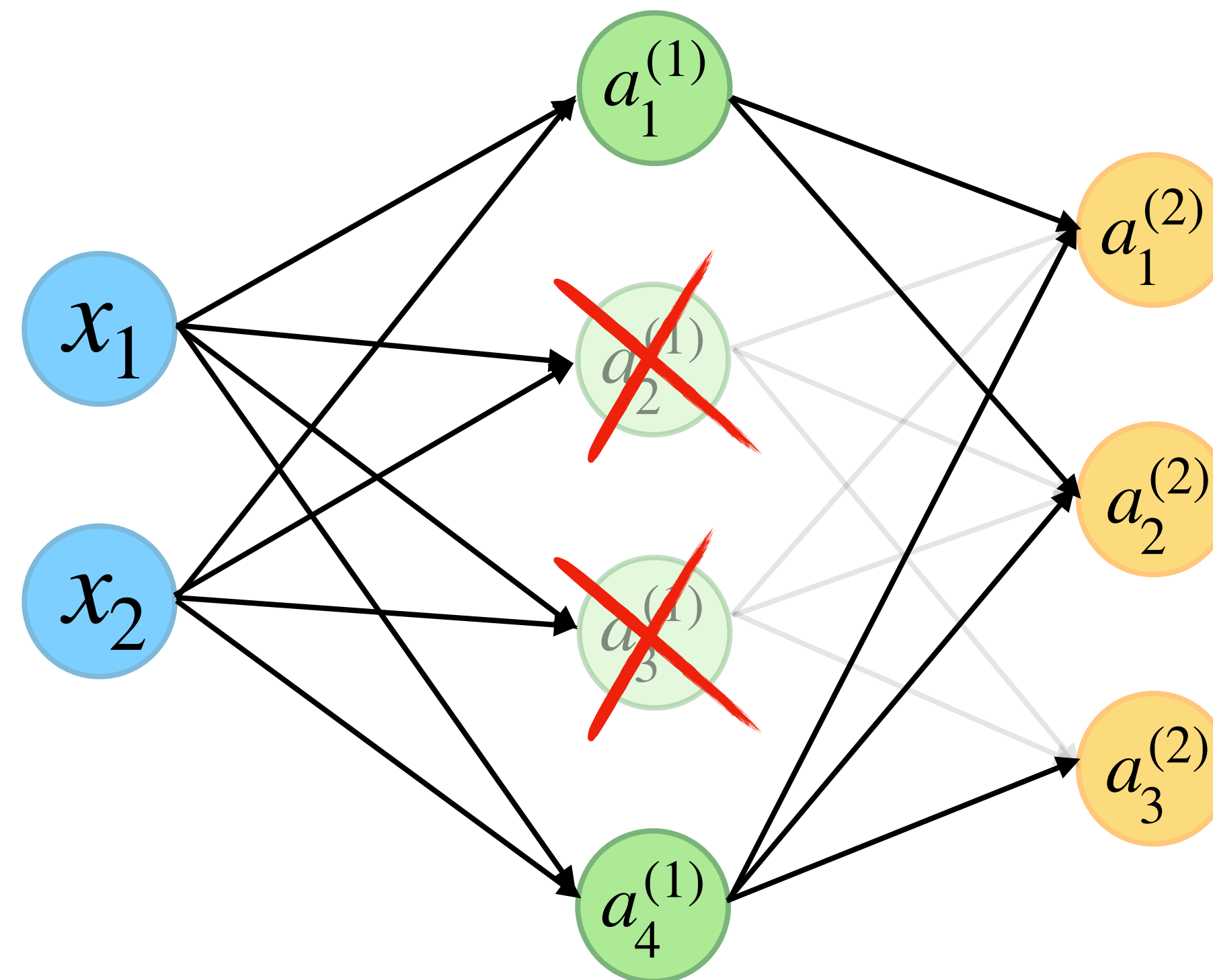


And during inference?

And **during inference?**

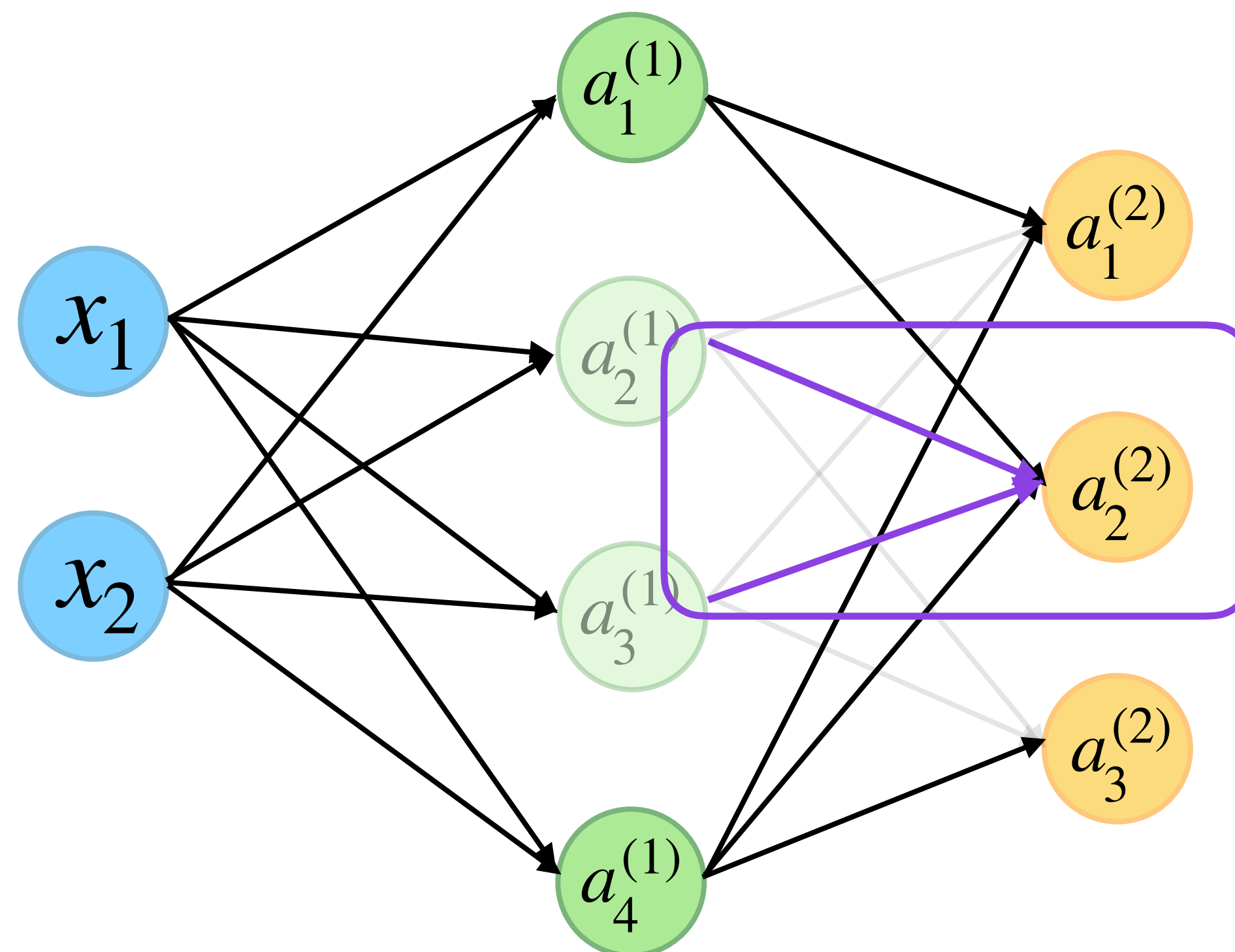
Deterministic behavior!
Don't drop nodes!

Note that activations are scaled during training



Assume we have
drop probability
 $p = 0.5$

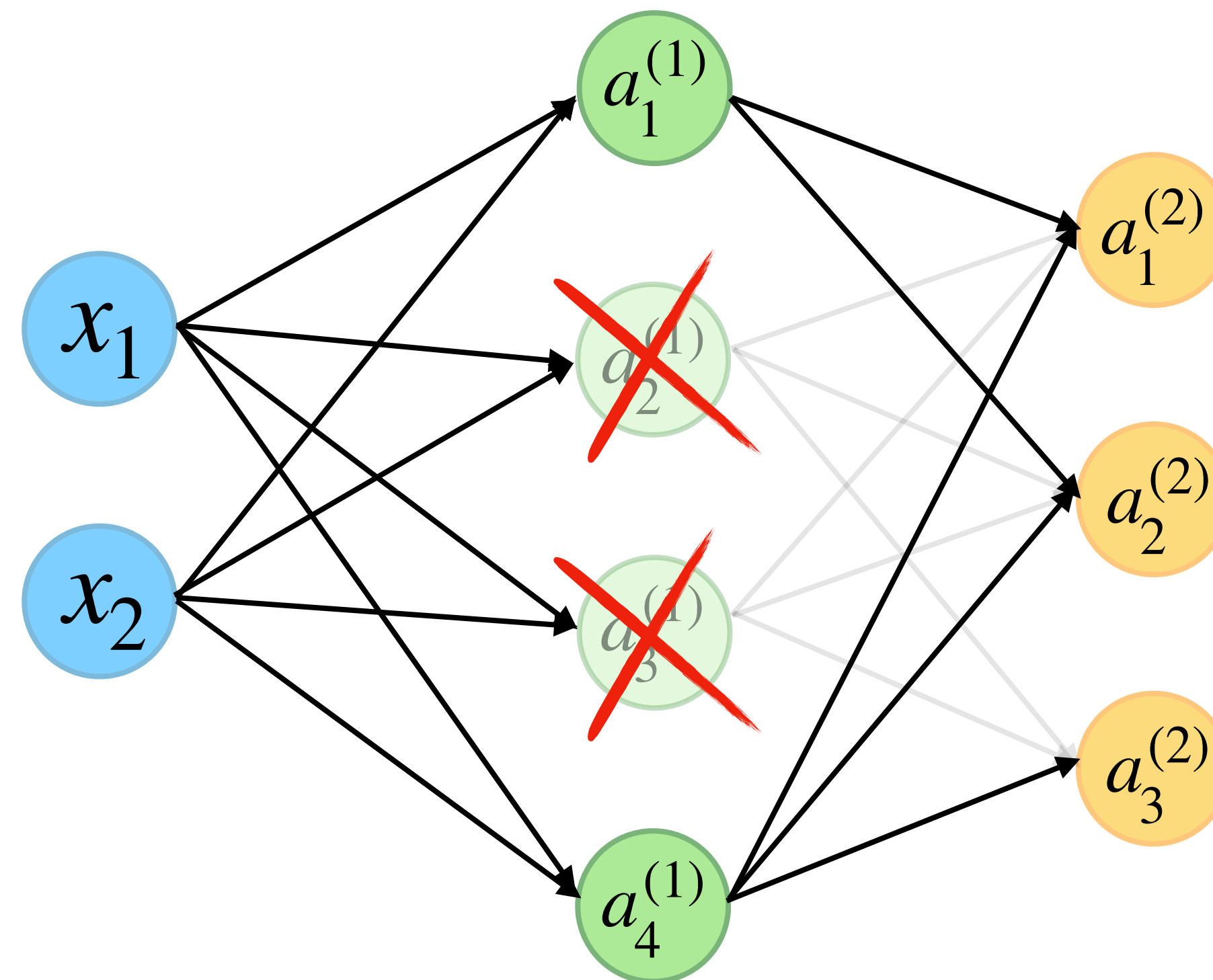
Note that activations are scaled during training



Assume we have
drop probability
 $p = 0.5$

During testing,
when nodes are
present again,
activations are
 $2 \times$ as large!

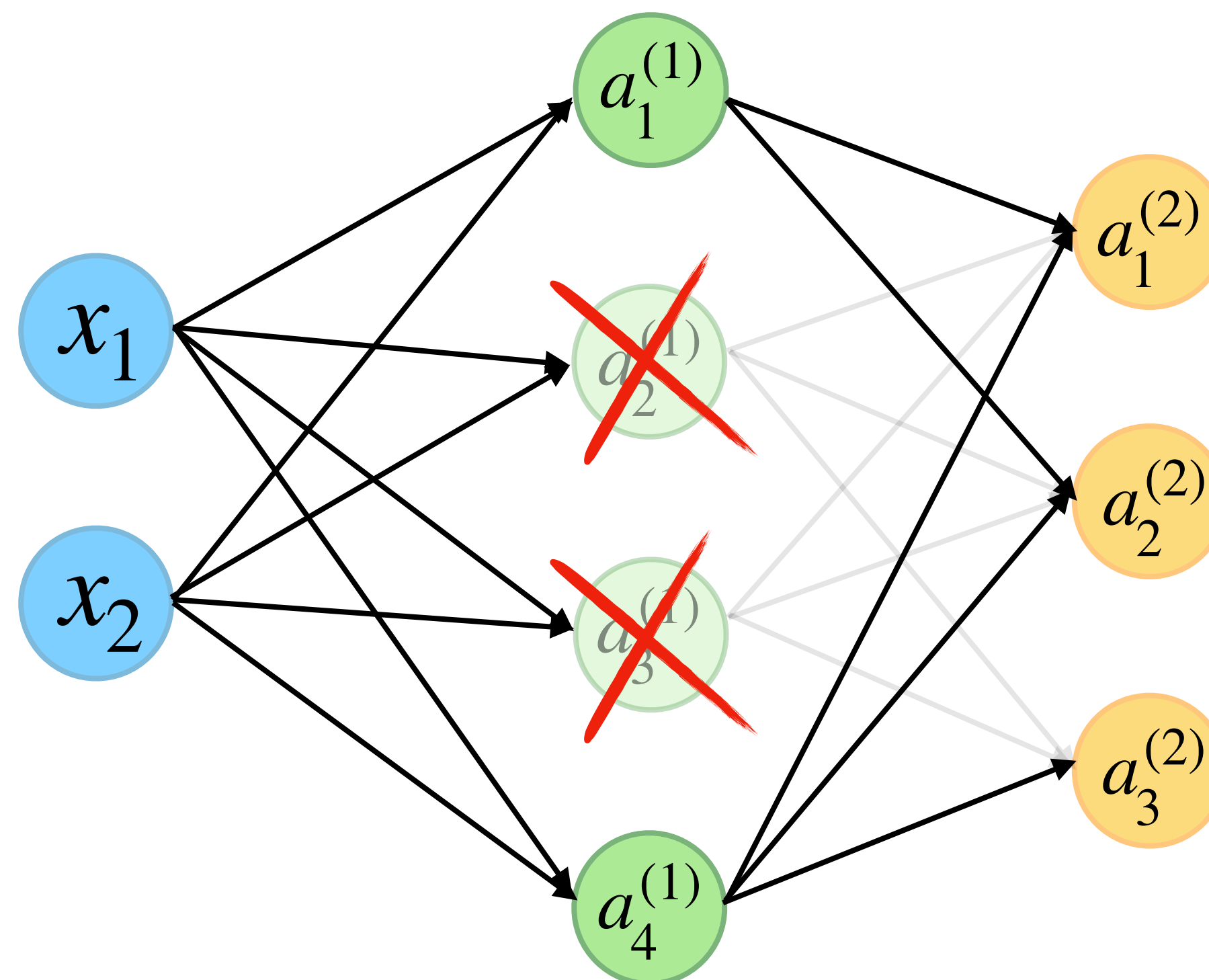
Note that activations are scaled during training



Assume we have
drop probability
 $p = 0.5$

During training,
node activations
are scaled by $1/p$

Note that activations are scaled during training



Assume we have
drop probability
 $p = 0.5$

During training,
node activations
are scaled by $1/p$

Example:

$$a = 5$$

$$a = 5/0.5 = 10$$

Next: Dropout in PyTorch